Exponents

An **exponent** is a shorthand notation for repeated multiplication of a number by itself.

$$a^n = a \cdot a \cdot a \ldots . \cdot a \ (n\ times)$$

where a is the base and n is the exponent.

Examples:

a) $10^3 =$ b) $2^5 =$

Notes:

i) $a^1 =$ ii) $a^0 =$ $(a \neq 0)$

3.1 Describe a decimal number in expanded notation

The decimal system is the most widely used number system. It is a base 10 system and uses digits 0 through 9.

Place Value

From right to left, the places are ones, tens, hundreds, thousands, ten thousands, and so on.

… 10 000 1 000 100 10 1

Each place value can be written as a power of 10:

$$1 = 10^0 \ ; \ 10 = 10^1 \ ; \ 100 = 10^2 \ ; 1\,000 = \ 10^3 \ ; 10\,000 = 10^4 \ ; \ \ldots$$

Moving from right to left, the exponent increases by one at each place.

3.2 Express a decimal number in expanded notation

To write a decimal number in expanded exponential notation, multiply each digit by its corresponding power of 10 and add the resulting terms.

Examples

Express $5\,427$ in expanded exponential form.

$5\,427 = 5 \times 10^3 + 4 \times 10^2 + 2 \times 10^1 + 7 \times 10^0$

Express $36\,891$ in expanded exponential form.

3.3 Describe a binary number in expanded notation

The binary system is a base 2 number system. The only digits used are 1 and 0.

A subscript 2 identifies a number as binary, for example 111_2 and 1101_2.

Place Value

From right to left, the places are ones, twos, fours, eights, sixteens, and so on.

… 16 8 4 2 1

Each place value is a power of 2:

$$1 = 2^0 \;;\; 2 = 2^1 \;;\; 4 = 2^2 \;;\; 8 = 2^3 \;;\; 16 = 2^4 \ldots..$$

In exponent form, the place values are

$$\ldots\ldots, 2^4, 2^3, 2^2, 2^1, 2^0.$$

To expand a binary number, start at the rightmost digit (2^0). Moving left, increase the exponent by one for each position. Then write the expansion from left to right.

3.4 Express a binary number in expanded notation

Examples

Expand 1101_2 as a sum of appropriate powers of 2.

- First, observe that the highest exponent place value in this number is 2^3.
- Expanding as with the Decimal System earlier, we get:

$$1\ 1\ 0\ 1_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 2^3 + 2^2 + 0 + 1$$

Since binary digits are either 0 or 1, we can simplify the expansion by including only the powers of 2 corresponding to digits that are 1.

Evaluating and adding these powers gives the decimal equivalent.

$$2^3 + 2^2 + 0 + 1$$

$$8 + 4 + 1$$

$$13_{10}$$

Binary Place-Value Reference

2^5	2^4	2^3	2^2	2^1	2^0
32	16	8	4	2	1

By first expanding using exponents, convert each binary number to its decimal equivalent.

a) $1\ 0\ 1\ 1\ 0_2$

b) $1\ 1\ 0\ 1\ 0\ 1_2$

c) $1\ 1\ 1\ 1\ 0\ 0\ 1_2$

Computer Applications

Computers use the binary number system to represent and store information. Binary uses only two digits, 0 and 1, which correspond to two possible electronic states.

Each binary digit (0 or 1) is referred to as a bit.

In computers, bits are commonly arranged in groups of eight.

Each group of eight bits is referred to as a byte.

Place value for 8-digit binary numbers

Bit positions are counted from right to left, beginning with the first position on the right.

Bit Location	8th	7th	6th	5th	4th	3rd	2nd	1st
Exponent								
Place Value								

Problems:

1. Identify the place value of the 1 in each byte.

   a) $0\,0\,0\,1\,0\,0\,0\,0_2$

   b) $0\,1\,0\,0\,0\,0\,0\,0_2$

2. Convert each binary representation to its decimal equivalent. Show your expansion.

   a) $0\,1\,0\,1\,0\,1\,1\,1_2$

   b) $1\,1\,0\,0\,1\,1\,0\,1_2$

3.9 Convert whole numbers using the following systems:

3.9.1 Decimal

3.9.2 Binary

Converting Decimal Numbers to Binary

Question: How would we convert a Decimal number to Binary form?

Answer: Determine which powers of 2 add up to the decimal number. Write 1 in the positions corresponding to powers of 2 that are needed, and 0 in the remaining positions.

Note

Every nonnegative integer has a unique binary representation when leading zeros are omitted. For example, 1101_2 and 00001101_2 represent the same value.

Express 13 in binary form.

Solution:

First, find the largest power of 2 less than or equal to 13. Since $2^4 = 16$ and $2^3 = 8$, the largest needed place value is 8.

Place Value	8	4	2	1
Digit	1			

Subtract 8 from 13 to get 5. Since 5 = 4 + 1, place a 1 in the fours and ones positions. The twos position is 0.

Place Value	8	4	2	1
Digit	1	1	0	1

Therefore, $13_{10} = 1101_2$.

Examples

Express each decimal number in binary form. Show your work.

a) 58

b) 118

c) 231

Larger Numbers

Question 1: What is the largest nonnegative decimal number that can be represented by an 8-bit unsigned binary number?

The largest unsigned 8-bit value occurs when all eight digits are 1: 11111111_2.
Expand using powers of 2, then evaluate:

Question 2: How could the decimal number 256 be represented in binary form?
$256 = 2^8$. This requires a ninth bit, so its binary representation has nine digits.

Question 3: What is the largest nonnegative decimal number that can be represented by a 9-bit unsigned binary number?
Begin with 111111111_2, then expand and evaluate.

Look for a pattern in your answers to Questions 1 and 3. The largest unsigned value represented by n bits is:

$$2^n - 1$$

Subsequent place values beyond 256 (2^8) are:

i) 2^9 = _____ (10-digit binary numbers)

ii) 2^{10} = _____ (11-digit binary numbers)

Examples

1. Convert the following binary numbers to decimal form.

a) $1\quad 0\quad 1\quad 1\quad 0\quad 0\quad 0\quad 0\quad 1\quad 1_2$

b) $1\quad 1\quad 0\quad 1\quad 0\quad 1\quad 0\quad 1\quad 0\quad 0\quad 1_2$

2. Express 375 in binary form.